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Decoding Video-Elicited Emotions from EEG

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TASK

Information

- Goal
  - Predict 1 of 6 emotions for each test trial (anger, disgust, fear, happy, neutral, sadness)
- Data
  - 31-channel EEG, 200 Hz, notch-filtered at 60 Hz
  - 336 training trials over 14 sessions
  - 24 test trials from 1 held-out session
  - Trials have different lengths, padded with inf
- Deliverables
  - submission.csv with one label per test trial
  - Runnable notebook
  - Short presentation
- **Test session is unseen at training time -> cross-session is the challenge**

TASK

Challenges

1 Small dataset

Only 336 trials / 6 classes. Deep models would overfit. We use feature-based pipelines.

2 Cross-session test

LOSO is the only honest validation: trials in the same session share drift, contact, and arousal.

3 Variable trial length

Trials range 50-460 s. Per-session statistics shift. We need length-invariant features.

4 No channel montage

Without electrode positions we cannot use spatial priors (DASM/RASM, frontal-alpha asymmetry).

Data overview

- Files
  - data.npz -> train (336, 31, 63800), test (24, 31, 63800)
  - trial_info_train.csv -> session, emotion_label, stim_start, stim_stop
- Class balance
  - 6 emotions, 56 trials each -> uniform
- Sessions
  - 14 training sessions (1, 3-15) + 1 held-out test session (likely session 2)
- Trial length
  - True length recovered by finding first inf-padded sample
  - Range ~ 50-460 s; very different across trials

```
Arrays in data.npz:
  train    shape=(336, 31, 63800)  dtype=float32
  test     shape=(24, 31, 63800)  dtype=float32

trial_info_train.csv: 336 rows, 6 cols
Columns: ['Unnamed: 0', 'session', 'trial_number', 'emotion_label', 'stim_start', 'stim_stop']
   Unnamed: 0  session  trial_number emotion_label    stim_start     stim_stop
0           0        1             0          fear  1.760639e+09  1.760639e+09
1           1        1             1         happy  1.760639e+09  1.760640e+09
2           2        1             2         anger  1.760640e+09  1.760640e+09
3           3        1             3       disgust  1.760640e+09  1.760640e+09
4           4        1             4         happy  1.760640e+09  1.760640e+09
```

```
Label distribution:
emotion_label
fear       56
happy      56
anger      56
disgust    56
neutral    56
sadness    56
Name: count, dtype: int64

Session distribution (14 sessions):
session
1     24
3     24
4     24
5     24
6     24
7     24
8     24
9     24
10    24
11    24
12    24
13    24
14    24
15    24
Name: count, dtype: int64

Labels per session:
emotion_label  anger  disgust  fear  happy  neutral  sadness
session
1                  4        4     4      4        4        4
3                  4        4     4      4        4        4
4                  4        4     4      4        4        4
5                  4        4     4      4        4        4
6                  4        4     4      4        4        4
7                  4        4     4      4        4        4
8                  4        4     4      4        4        4
9                  4        4     4      4        4        4
10                 4        4     4      4        4        4
11                 4        4     4      4        4        4
12                 4        4     4      4        4        4
13                 4        4     4      4        4        4
14                 4        4     4      4        4        4
15                 4        4     4      4        4        4

Trial length (s): train [51.4, 318.5], test [48.1, 303.8]
```

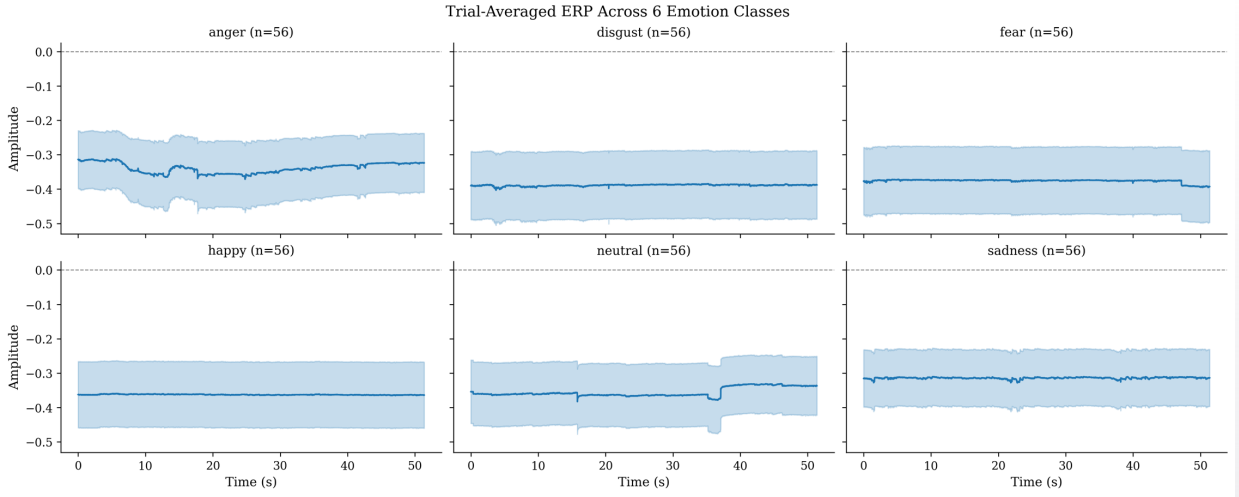
Trial length mismatch

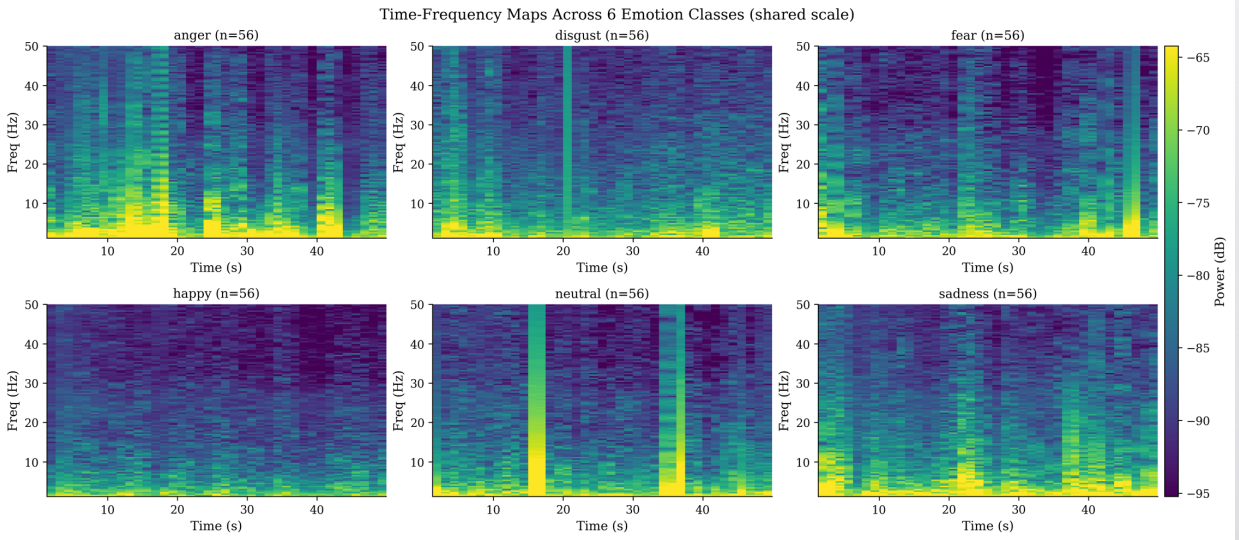
- Two ways to estimate trial length:
- From .npz signal
  - count non-inf samples
- From metadata
  - stim_stop − stim_start
- For some trials, metadata says it is longer than the actual signal
- **Fix**
  - usable_length = min(non_inf_length, stim_duration_length)

```
stim_duration vs npz length (s):
       stim_duration  npz_length
count         336.00      336.00
mean          140.07      136.81
std            62.38       57.84
min            51.37       51.37
25%            94.05       94.05
50%           126.02      124.21
75%           173.41      168.49
max           460.20      318.50
```

ERP and Time-Frequency maps

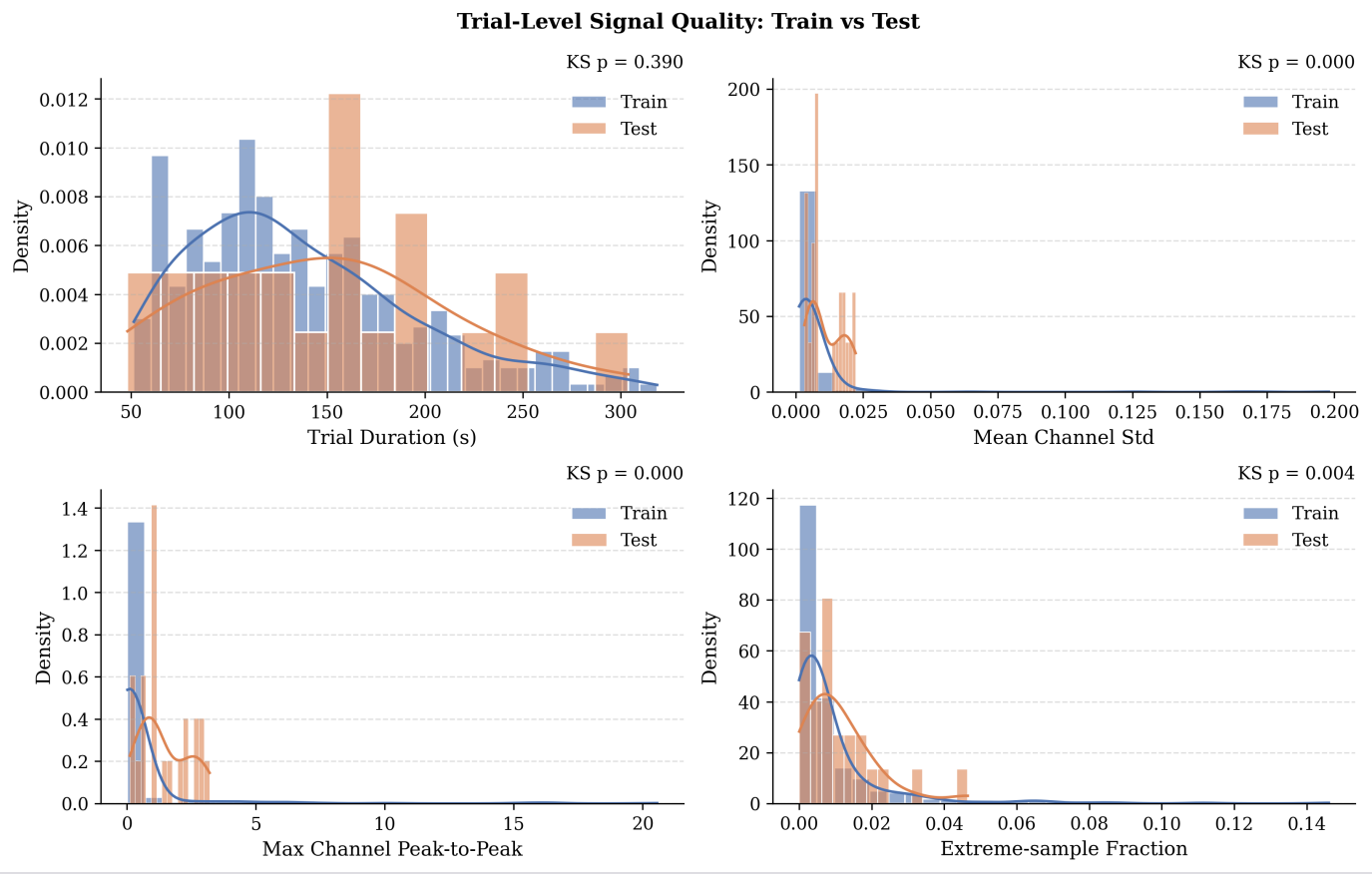
- ERP (trial-average across 7 channels)
  - All 6 classes overlap, no class-specific shape
  - -> no phase-locked signal, ERP features are useless
- Time-Frequency (Welch, 512 win, 50% overlap)
  - Most class differences live below ~10 Hz
  - Vertical broadband bursts = motion / EMG artifacts
- **Decision**
  - Work in the frequency domain
  - Apply high-pass filter, edge crop, robust normalization





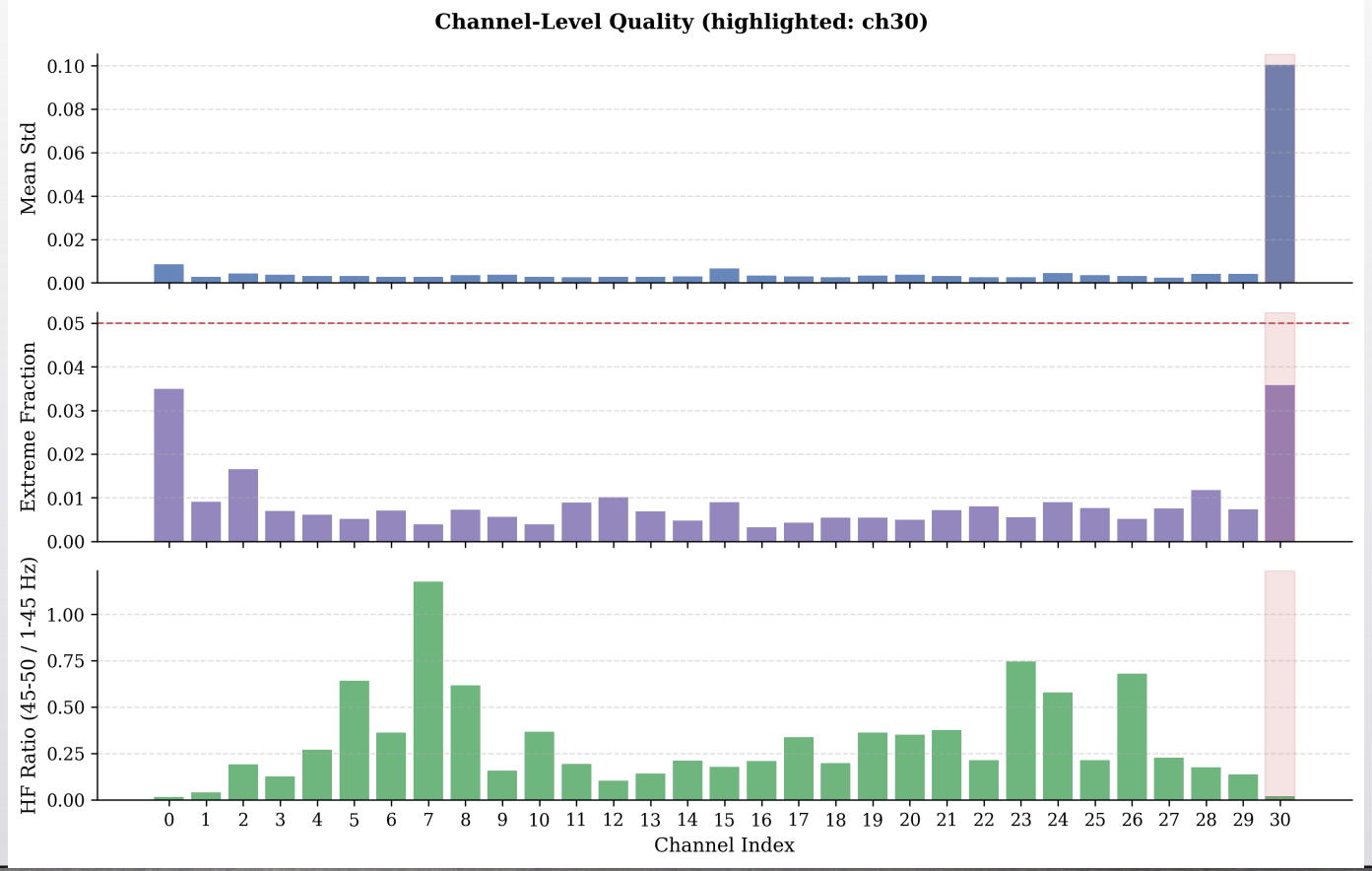
Trial level: train vs test

- Kolmogorov-Smirnov test on 4 metrics
  - duration, mean std, peak-to-peak, extreme-sample fraction
- Findings
  - Duration matches train and test (p = 0.39)
  - Other 3 metrics differ significantly (p < 0.005)
  - Test session has bigger amplitude scale than training
- **Solution**
  - Per-session robust z-score (train side)
  - CORAL alignment at test time (covariance match)



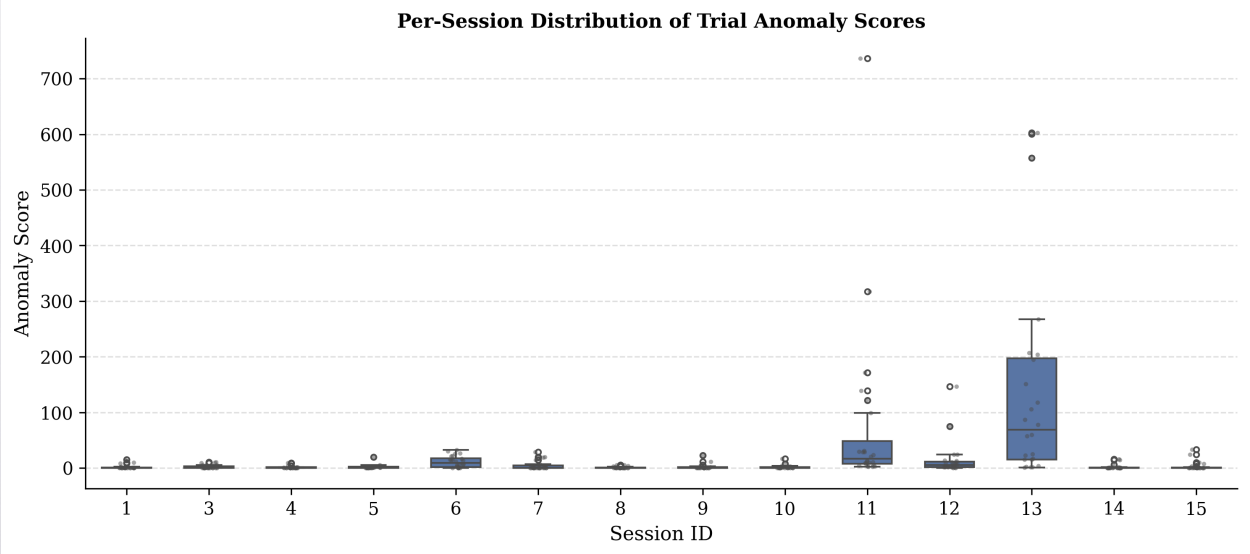
Channel level

- Per-channel metrics
  - mean std, extreme fraction, HF ratio (45-50 / 1-45 Hz)
- **Channel 30**
  - Std 10x higher than the rest, no high-freq energy
  - -> low-frequency drift, not EMG. Drop it.
- Channels 5-8 and 23-26
  - Higher HF ratio (frontotemporal EMG)
  - Flagged but kept (no LOSO improvement when dropped)



Session level: anomaly score

- Anomaly score per trial
  - z-score 3 stats (mean std, peak-to-peak, extreme fraction)
  - Sum the positive z-scores
- Findings
  - Most sessions cluster near zero
  - Sessions 11 and 13 have inflated scores
  - 4 trials in session 13 score above 1000
- **Decision**
  - Don't drop trials - robust per-session normalization absorbs them



Cleaning pipeline

1

Drop ch30

Bad channel — low-freq drift only

2

Crop padding

Use min(non_inf_length, stim_duration)

3

Edge crop

Trim 3 s front (emotion latency) + 1 s back

4

Detrend + 0.5 Hz HP

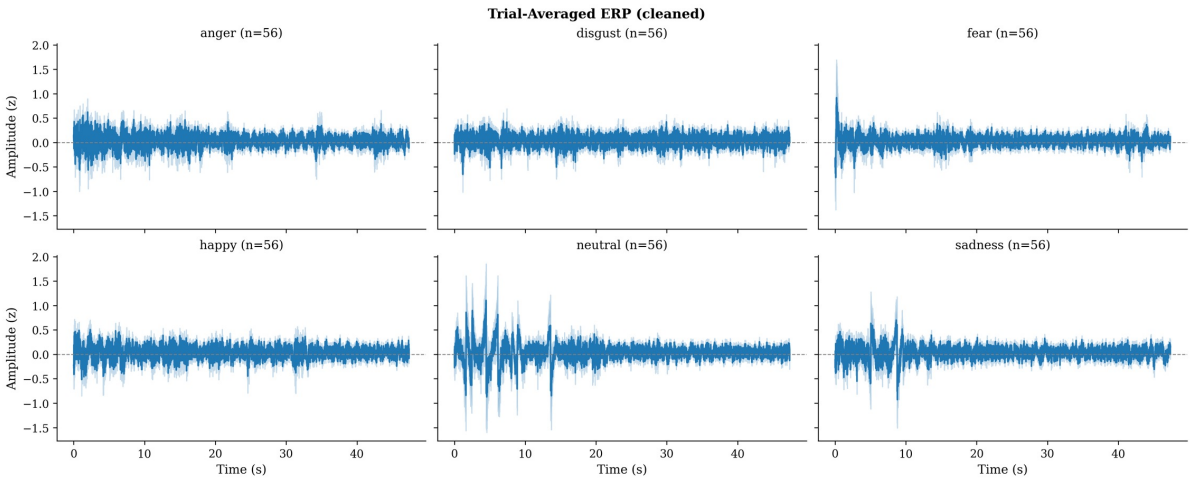
Linear detrend, zero-phase Butterworth

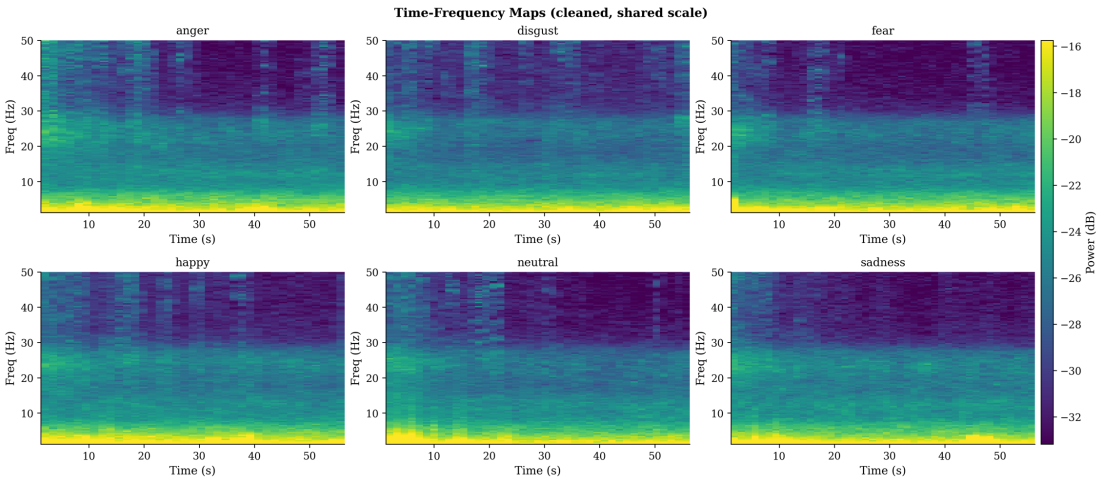
5

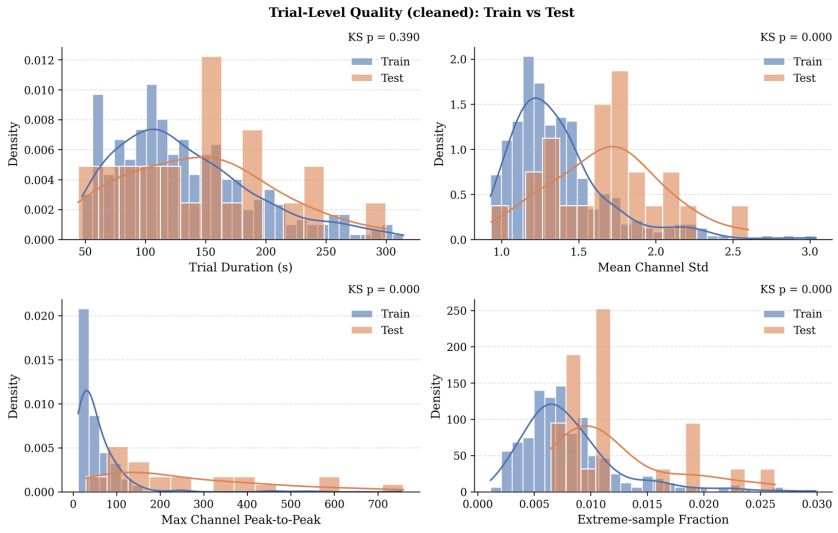
Per-session z

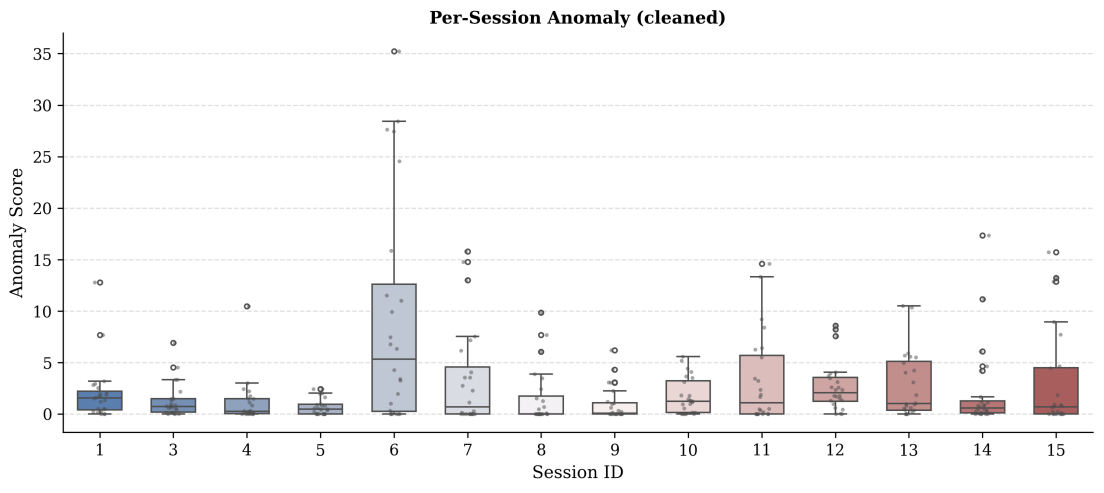
Robust z-score (median / MAD) within each session

After cleaning: visual + audit









Features

- Why frequency-domain features
  - Invariant to channel order and trial length
  - ERP showed no phase-locked signal
- Sliding window: 8 s, 50% overlap
  - Each window gives one feature vector
  - Aggregate windows -> mean, std, median per trial
- Per window, per channel:
  - Band power x 5 (delta, theta, alpha, beta, low gamma)
  - Differential entropy x 5
  - Hjorth parameters x 3 (activity, mobility, complexity)
- **Total per trial: (5+5+3) × 30 ch × 3 stats = 1170 features**

Feature counting

Per window

band power × 5
diff. entropy × 5
Hjorth × 3
total = 13 × 30 ch = 390

Per trial

3 stats × 390
= 1170 features

Classifier + ensemble

- Two heads
  - Spectral head: Logistic Regression on 1170 features
  - Riemannian head: covariance -> tangent space -> Logistic Regression
- Combine in log-prob space
  - log p = w · log p_spec + (1 - w) · log p_riem
  - Sweep w over 0.5-1.0, pick best by LOSO
- Class-prior balancing (per session)
  - Soft: Sinkhorn-style, push class proportions toward uniform
  - Hard: Hungarian assignment, exact 4 trials per class
- **Why balancing helps**
  - Test session is likely known to be balanced (4/class)
  - LR predictions drift toward majority - balancing corrects it

LOSO ablation (14 sessions)

#	Pipeline	Mean	Std
1	Baseline (raw, full band power)	0.452	0.132
2	+ Cleaning (drop ch30)	0.482	0.127
3	+ Stim crop + head/tail trim	0.503	0.113
4	+ Per-session z + CORAL	0.593	0.110
5	+ Windowing + DE + Hjorth	0.637	0.111
6	+ Riemannian ensemble, soft (w=0.9)	0.646	0.125
7	+ Riemannian ensemble, hard (w=0.9)	0.639	0.139
8	**Final (soft, best w)**	**0.655**	**0.130**
9	**Final (hard, best w)**	**0.652**	**0.135**

BEST

0.655

± 0.130 (LOSO)

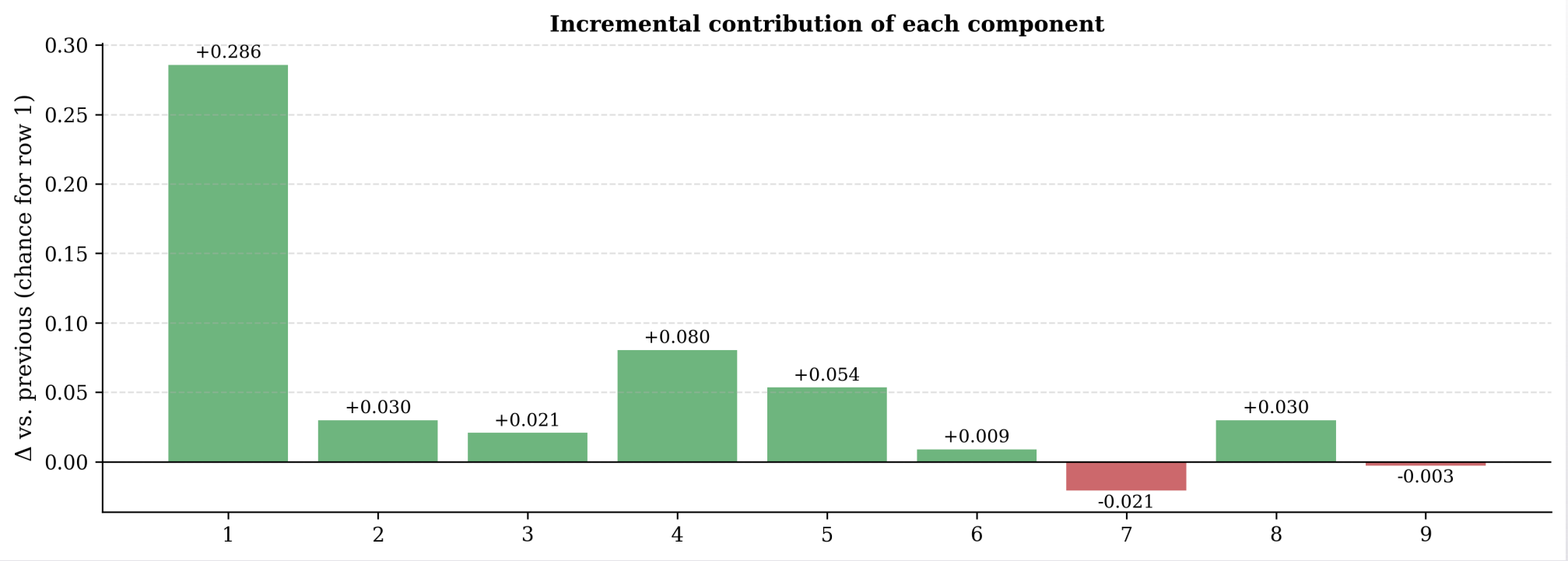
chance = 0.167

+0.22 over baseline

Submitted: Row 9 (hard assignment) - exact balanced predictions

Rows 8-9: best ensemble weight chosen by LOSO sweep

Where the gains come from



CORAL alignment is the single biggest jump (rows 3 → 4)

What we would do next

01

Get the channel montage

Unlocks spatial priors: DASM, RASM, frontal-alpha asymmetry, hemispheric lateralization. These drive most published affective-EEG pipelines.

02

More sessions per subject

Reduces LOSO variance and lets us learn session-invariant representations directly (e.g. domain-adversarial training).

03

Tune window size + features

Window length, hop, band edges, and DE estimator are all heuristic. A clean grid search with proper LOSO would help.

04

Validate emotion-onset latency

The 3 s front-trim is a guess. Neuroscience and psychology work, or a small pilot, could tell us when each emotion really starts to change.

Thank you

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